



Galaxy S10+ exclusive skin for Fortnite

Samsung has made an exclusive K-pop-themed skin for Fortnite players who buy the new Galaxy S10+ phone. <https://digit.in/mar19-36>



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Android devices running Google Play Services 7.0 and later can use your fingerprint or PIN to log in <https://digit.in/mar19-37>

may be captured during this time. The multiple exposures captured are individually of a longer duration than the exposures captured with HDR+. Feedback from the gyroscope is used to eliminate exposures that may be shaky. Night Sight is a dedicated mode, and cannot be used with other modes such as portrait or panorama. If the phone senses low light conditions, it prompts the user to turn the mode on.

The iPhone XS Max has a feature known as Smart HDR. At the press of the button, the phone quickly combines multiple exposures into a single photo, and the process is nearly instantaneous, as the phone has already captured the image before you tap the button. The magic of the Smart HDR feature really shines in the video mode, where the multiple exposures are combined in real time, and the user actually sees the end result while shooting. In the video mode, the feature extends the dynamic range only up to 30 fps, and not for higher frame rates.

BETTER PORTRAITS

The camera is the fastest evolving components of smartphones, and there can be big jumps in just a single generation. There is actually a huge difference in the capabilities of the iPhone X and the iPhone XS Max, when it comes to accurate colour representation, how blown out portions of an image can be, and focusing speed. Another major improvement is the bokeh effect. Depth control on the XS Max allows users to change the f-number, or focusing distance after the image has been clicked. This allows users to control the amount of background blur, to create images where the subject pops. The XS Max supports a range between f16 to f1.4. This allows users to control the amount of the bokeh effect, with the end result being closer to what can be expected from DSLR cameras.

In the latest generation of Apple devices, the neural engine kicks in as soon as the phone is put in the portrait mode. Machine learning algorithms identify the faces in the frame. The Vision framework then “landmarks” the face, identifying the eyebrows,

eyes, nose, lips and the contours of the face. This information is used to tweak the lighting. The depth engine and the neural engine then separate the subject from the background, drawing a mask. Now comes the fun bit, there are five lighting options, Natural Light, Studio Light, Contour Light, Stage Light, and Stage Light Mono. Studio Light boosts the brightness of the face, Contour Light adds natural looking shadows depending on the shape of the face, and the last two options completely remove the background. The lighting effects can be selected while capturing an image, and even after the image has been captured, as long as it was taken in the portrait mode. The Honor 10 offers similar portrait lighting modes as well, including the face isolation in modes, which are called Theater Light and Classic Light.

Samsung uses a similar approach to isolate humans in shots, then allowing users to change the depth of field in a feature called Live Focus. In the S10+, the S10 and the S10e, users can pick between four types of bokeh effect. These are Blur, Spin Bokeh, Zoom Bokeh and Colour Point. These are essentially radial blur and zoom blur options to lead the eyes to the subject of the photo. The Colour Point effect changes the background to black and white.

The Google Pixel 3 manages to provide the bokeh effect using a single camera setup, by relying on AI and phase detection autofocus (PDAF). In the Pixel 2, the camera would capture two images from slightly different angles, and isolate the human subject based on the differences between the two images on a pixel level. While the subject remains stationary, the background moves slightly, in a horizontal direction if the camera is held horizontally, and in a vertical direction if the camera is held vertically. This informa-

tion is then used to create a mask that isolates the subjects, and blurs the background. As the depth information is not available on the lines parallel to the line of sight of the camera, this was not a perfect process, and at times, there could be strange aberrations in the images. The Pixel 3 uses another depth cue from PDAF, where the sharpness of background is also considered. Objects that are closer to the camera are sharper. An additional depth cue

is that the rough size of the human head is known, and by counting the number of pixels that the face occupies, the smartphone can figure out how far away the person is. The depth estimation on the Pixel 3 uses a neural network trained to create depth masks. The training required a large number of PDAF images, as well as high quality

depth masks. These were produced by a custom rig called the Frankenphone, which used five synchronised Pixel 3 cameras. Once the depth mask is created, it is easy to adjust the amount of background blur. The system has not just been trained on faces, but also other objects as well. One of the best results are for photographs of flowers.

ARTIFICIAL INTELLIGENCE

AI plays a major role in capturing and processing HDR and portrait shots. Smartphone manufacturers use neural networks and machine learning algorithms in various ways to help users capture better photographs, and then process them.

The latest Samsung Galaxy S10 lineup introduces a feature known as Shot Suggestions, a feature that uses AI that has been trained on over 100 million high quality professional photos. This optional feature is a guide for better composition, and uses a white line and a dot to help you better frame your



The Frankenphone rig used for depth map training

CREDIT: GOOGLE AI BLOG